

Governed Autonomy as a Manufactured Artifact

A Mathematical Theory of Certified Allocation, Its Deletion of
Implementation, and the Geometry of the Phase Transition
it Induces in Autonomous Software Agents

Sean Chatman
AutoFDE Laboratory
sean@chatmangpt.com

September 2026

Abstract

Autonomous agents must allocate finite resources across exploration options, and every classical policy family (greedy, ε -greedy, UCB1, PUCT) optimizes performance while remaining structurally incapable of governance: outputs are non-deterministic across platforms, non-conserving under discretization, silently degrading under failure, and invisible to audit. This dissertation develops, in full formal detail, the mathematics of the alternative demonstrated by the AutoFDE artifact chain: allocation as a *certified, receipted, host-invariant artifact*.

The mathematical contributions are layered. *Algebra*: the escort cascade is developed through its partition function $Z(q) = \sum_i m_i^q$, Rényi dimensions D_q , and the Legendre duality between the lens exponent q and the local Hölder spectrum $f(\alpha)$; the substitution of integer-lens powers m^q , $q \in \mathbb{Z}$, for the transcendental exponential $e^{\tau S}$ is proved to algebraize determinism—bit-identical replay across CPython and BEAM hosts becomes a theorem about the dyadic grid $\mathbb{D}_{16} = \mathbb{Z}/2^{16}$ rather than a hope about floating point. *Geometry*: allocations live in the simplex Δ^{N-1} ; canonical hash-ordering exhibits the allocator as a function on the orbit space $\Delta^{N-1}/\mathfrak{S}_N$; the compiled policy is measured to have the exactly-uniform symmetric fixed point, to *fund rarity* (a near-zero-mass candidate receives 0.1721, more than the minimum strong share 0.1183, via the rare lens $q = -1$), and to compress four orders of magnitude of mass ratio into a factor 3.4 of share ratio. A conjectured coverage floor of $\Lambda_0/N = 5/32$ is *refuted by measurement* (raw weak share 0.0837), and the refutation is retained as a first-class result: the ρ -routed, multiplicatively-weighted cascade is not a barycenter of its lenses, and its sharp floor is stated as an open problem. *Calculus*: the lens derivative $\frac{d}{dq} \log Z(q) = \mathbb{E}_{L_q}[\log m]$ and the Gibbs identity $\frac{dH}{d\tau} = -\tau \text{Var}(S)$ are proved; the sum-product law of combinatorial maximalism is formalized (option graphs grow with the sum of alternatives while path cardinalities carry their product), giving the information-theoretic reason governance must preserve option entropy rather than maximize point yield. The phase transition itself is given quantitative form: per-implementation governance has integrity $(1-p)^H$, exponential in hosts H , while the single-kernel law replaces it with a fixture of constant cost—the availability argument of the transistor, transposed from components to hosts.

The dissertation’s method is its own subject: a falsification ledger in which the circular benchmark, the every-softmax “theorems,” the notational equivalence proof, the local reference implementation, and most recently the coverage-floor conjecture were each deleted or refuted with commit-anchored receipts. What survives is the governance envelope and its phase-change structure, argued with explicit disanalogies and falsifiers.

Contents

I	The Program	3
1	Introduction	4
1.1	The problem	4
1.2	The thesis statement	4
1.3	Contributions	4
1.4	Reading discipline	5
2	Background and Related Work	6
II	The Mathematical Development	7
3	The Geometry of Governed Allocation	8
3.1	The simplex, the symmetric group, and the orbit space	8
3.2	The budget polytope and its lattice geometry	9
3.3	The floor projection: apportionment theorems	9
3.4	Lanes, depth, and gauge structure	10
4	The Escort Cascade: Algebra and Multifractal Calculus	11
4.1	Masses and the partition function	11
4.2	Rényi dimensions and the lens calculus	11
4.3	The Gibbs identity for the deleted reference engine	12
4.4	Legendre duality: the Hölder spectrum	12
4.5	Information geometry of the escort family	13
5	The Compiled Policy: Emergent Geometry of the Certified Blend	14
5.1	The symmetric fixed point	14
5.2	A conjectured coverage floor, refuted	15
5.3	Rarity funding: the measured anti-starvation mechanism	15
5.4	A decision-theoretic reading (model)	16
6	Arithmetic Determinism: Dyadic Rationals and the Transcendental Bound- ary	17
6.1	The dyadic grid	17
6.2	What IEEE 754 does and does not promise	17

6.3	The algebraization theorem	17
6.4	The executed proof: the cross-host golden fixture	18
6.5	Determinism of the section	18
7	Conformance Geometry and Adversarial Separation	19
7.1	Event data as typed multigraphs	19
7.2	Alignments as geodesics	19
7.3	The counterfactual battery as adversarial neighborhood	19
8	Combinatorial Maximalism: The Sum–Product Law of Option Preservation	21
8.1	The sum–product law	21
8.2	Option entropy as log-volume	21
8.3	Pareto exposure, not argmax	21
8.4	Reversibility lattice	22
9	The Phase Transition, Formally	23
9.1	Availability functionals	23
9.2	Failure-class removal as language emptiness	23
9.3	The order parameter	23
III	Evidence and Argument	25
10	The Governance Envelope, Summarized	26
11	The Falsification Ledger	27
12	The Phase-Change Analysis	28
12.1	A model of phase change	28
12.1.1	Steam	28
12.1.2	The transistor	28
12.2	The mapping	28
12.3	Disanalogies, stated	28
12.4	Falsifiers	29
13	Conclusion	30
A	Artifact and Verification Map	31
B	Compiled Constants	32
C	Measured Fixtures	33
D	The Deleted Claims, For the Record	34

Part I

The Program

Chapter 1

Introduction

1.1 The problem

Let an autonomous agent face a finite frontier of candidate branches $c_1, \dots, c_N$, each characterized by option entropy $\Omega_i \geq 0$, historical yield Y_i , and cost $C_i > 0$, and let it hold a discrete budget $B = (B_{\text{tick}}, B_{\text{mem}}, B_{\text{depth}}, B_{\text{risk}})$. The allocation problem is the map

$$A : (\text{features}, B) \mapsto \mathbf{s} \in \Delta^{N-1} \mapsto \mathbb{Z}^N \cap B \cdot \Delta^{N-1}, \quad (1.1)$$

from real-valued evidence to a point of the probability simplex and thence, by discretization, to integer allocations. Every classical policy instantiates (1.1); none of them makes the *decision artifact*—the output $\mathbf{s}$ and its integer projection—auditable, replayable, or conserved. This dissertation develops the mathematics of the ones that do, and argues that doing so is a phase transition in the sense of Chapter 9.

1.2 The thesis statement

Principle 1.1 (Thesis statement). When the map (1.1) is manufactured as a single certified artifact—deterministic across hosts (Ch. 6), conserving by construction (Ch. 3), refusing by type rather than degrading (Ch. 10), and emitting content-bound receipts into process event data (Ch. 7)—the failure class of divergent, silent, unverifiable implementation is removed structurally, and composition across hosts and trust boundaries becomes arithmetically possible rather than merely permitted.

1.3 Contributions

1. **Geometry (Ch. 3)**: the orbit-space formulation of deterministic replay; the lattice geometry of the budget polytope with its Ehrhart count; and four apportionment theorems for the floor projection (feasibility, monotonicity, residual bound, permutation equivariance), connecting budget conservation to the classical apportionment problem of Balinski and Young.

2. **Algebra and multifractal calculus (Ch. 4)**: the escort cascade through $Z(q)$, D_q , $\tau(q)$, and the Legendre duality with the Hölder spectrum; the lens derivative identity; the Gibbs entropy identity; and the identification of the escort family with the α -geometry of information geometry.
3. **Emergent policy geometry (Ch. 5)**: the exactly-uniform symmetric fixed point (verified), the rarity-funding phenomenon (measured), extreme mass-ratio compression (measured), and the refutation of the natural coverage-floor conjecture, with the sharp floor stated as an open problem.
4. **Arithmetic determinism (Ch. 6)**: the dyadic-grid model of Q16.16; the algebraization theorem replacing $e^{\tau S}$ by m^q with $q \in \mathbb{Z}$; and the bit-identity theorem with its executed proof (the cross-host golden fixture).
5. **Conformance geometry (Ch. 7)**: alignments as geodesics of the synchronization product; the counterfactual battery as an adversarial neighborhood with a separation condition.
6. **Combinatorial maximalism (Ch. 8)**: the sum-product law of option preservation; option entropy as log-volume; support invariants; and the Pareto-exposure discipline.
7. **The phase transition, formally (Ch. 9)**: availability functionals $(1 - p)^H$ versus constant-cost verification; failure-class removal as language emptiness; the hand-written fraction as an order parameter with a monotone descent law.
8. **The falsification ledger (Ch. 11)**, including a conjecture refuted during the writing of Chapter 5.

1.4 Reading discipline

Theorems are proved, propositions are proved or verified exactly, measurements are labeled *measured* and reproduced from Appendix A, and model-level arguments (Bayesian hedging, availability asymptotics) are labeled *model* and their assumptions stated. Nothing in Part II is decoration: each formal object is the mathematical shadow of a deployed artifact, and where the formalism failed audit, the failure is printed.

Chapter 2

Background and Related Work

Allocation under uncertainty. UCB1’s logarithmic regret [2], PUCT search [20], and the general theory [22] characterize the performance frontier under a trust model in which the policy’s implementor is trusted absolutely. This dissertation inverts the trust model and asks what the decision artifact must satisfy for third-party replay.

Information geometry and multifractal analysis. The escort distributions $L_q(i) \propto m_i^q$ originate in the multifractal formalism [8, 18] and in nonextensive statistics [23]; their geometric home is Amari’s α -geometry [1], which we connect to the lens parameter in §4.5. Entropy variational principles follow Jaynes [10] and Shannon [21]; elementary facts follow Cover and Thomas [6].

Apportionment. Discretizing quotas onto integer budgets is the classical apportionment problem; the failures of naive rounding methods are catalogued by Balinski and Young [3]. The floor projection of Chapter 3 is deliberately the simplest member of this family, and its theorems are the corresponding weakest (hence most robust) guarantees.

Lattice geometry. Counting lattice points in dilated polytopes is Ehrhart theory; the standard reference is Beck and Robins [5].

Floating point. IEEE 754 specifies correct rounding for $+$, $-$, $\times$, $\div$ and leaves transcendental functions to implementations; the consequences are classical [7, 9]. Chapter 6 turns this negative fact into the positive design decision that makes bit-identity purchasable.

Object-centric process mining. OCEL 2.0 [15, 17] and object-centric process querying [16] supply the durable event substrate into which receipts are projected (Chapter 7).

The industrial analogues. Landes [11], Rosenberg [19], Mokyr [13] establish the constraint-class-removal model of general-purpose transitions; Bardeen and Brattain [4] and Moore [14] supply the transistor instance; MacKenzie [12] the sociology of mechanized trust.

Part II

The Mathematical Development

Chapter 3

The Geometry of Governed Allocation

3.1 The simplex, the symmetric group, and the orbit space

Fix the compiled shape $N \leq 8$. Write

$$\Delta^{N-1} = \left\{ \mathbf{s} = (s_1, \dots, s_N) \in \mathbb{R}_{\geq 0}^N : \sum_i s_i = 1 \right\} \quad (3.1)$$

for the standard simplex, a compact convex polytope with N vertices $e_1, \dots, e_N$. The symmetric group $\mathfrak{S}_N$ acts on Δ^{N-1} by permuting coordinates, $\sigma \cdot (s_1, \dots, s_N) = (s_{\sigma^{-1}(1)}, \dots)$, and the physically meaningful state of an *unordered* frontier is a point of the orbit space

$$\Delta^{N-1}/\mathfrak{S}_N = \left\{ \text{nonincreasing rearrangements} \right\} = \left\{ s_1 \geq s_2 \geq \dots \geq s_N \geq 0, \sum s_i = 1 \right\}, \quad (3.2)$$

the *Weyl chamber* of type A_{N-1} : a closed simplex glued from $N!/2$ fundamental domains along reflection walls.

Definition 3.1 (Canonical section). The *canonical hash section* is the map κ assigning to each candidate c_i the digest $\kappa(c_i) = \text{SHA-256}(\text{blob}(c_i))$ and ordering candidates by $(\kappa(c_i), c_i)$ lexicographically; the induced section $\sigma_\kappa : \Delta^{N-1}/\mathfrak{S}_N \rightarrow \Delta^{N-1}$ selects, in each orbit, the representative whose coordinates are indexed by canonical order.

Proposition 3.2 (Determinism as descent to the quotient). *Suppose the measure map $M : \mathcal{C}^N \rightarrow \Delta^{N-1}$ satisfies $M(c_{\sigma(1)}, \dots, c_{\sigma(N)}) = \sigma \cdot M(c_1, \dots, c_N)$ ($\mathfrak{S}_N$ -equivariance) and is computed by arithmetic that is exact on its value grid. Then $M \circ \sigma_\kappa$ is a well-defined function on the orbit space, and identical inputs yield bit-identical outputs.*

Proof. Equivariance means M is constant on orbits up to the action; the section selects a unique representative; exactness on a totally ordered grid removes implementation latitude. Uniqueness of the representative forces uniqueness of the output bytes. $\square$

Remark 3.3. Proposition 3.2 is the formal content of the “canonical sort” engineering rule: sorting by hash is not a convenience but the choice of a measurable section for a group action, and replay determinism is the statement that the composite descends. The golden fixture of §6.4 is its executed proof.

3.2 The budget polytope and its lattice geometry

Discrete allocations live in the dilated simplex

$$B \cdot \Delta^{N-1} = \left\{ x \in \mathbb{R}_{\geq 0}^N : \sum_i x_i \leq B \right\}, \quad B \in \mathbb{N}, \quad (3.3)$$

and the feasible integer allocations are the lattice points $B \cdot \Delta^{N-1} \cap \mathbb{Z}^N$.

Proposition 3.4 (Ehrhart count of the budget simplex). $|B \cdot \Delta^{N-1} \cap \mathbb{Z}^N| = \binom{B+N}{N}$.

Proof. Lattice points are weak compositions $x_1 + \dots + x_N + x_{N+1} = B$ in $N+1$ nonnegative parts (the slack variable absorbs the inequality); the count is stars and bars, the degree- N Ehrhart polynomial of the simplex. $\square$

Remark 3.5 (Maximalism, first appearance). The policy does not and must not enumerate $\binom{B+N}{N} \sim B^N/N!$ allocations (for $B = 9973$, $N = 8$ this exceeds 10^{28}); it selects one point of this lattice by a deterministic rule. Chapter 8 elevates the discipline: *bounds, not enumerations*.

3.3 The floor projection: apportionment theorems

Let $\mathbf{s} \in \Delta^{N-1}$ with active set $A = \{i : s_i \geq \theta\}$, $\sum_{i \in A} s_i = 1$ after renormalization. The deployed projection assigns $\lfloor s_i B \rfloor$ of each discrete budget dimension.

Theorem 3.6 (Floor apportionment). *The map $\Phi_B : \mathbf{s} \mapsto (\lfloor s_i B \rfloor)_{i \in A}$ satisfies:*

- (i) (Feasibility) $\sum_{i \in A} \lfloor s_i B \rfloor \leq B$;
- (ii) (Residual bound) $B - \sum_{i \in A} \lfloor s_i B \rfloor = \sum_{i \in A} \{s_i B\} < |A|$;
- (iii) (Monotonicity) $s_i \geq s_j \Rightarrow \lfloor s_i B \rfloor \geq \lfloor s_j B \rfloor$;
- (iv) ($\mathfrak{S}_N$ -equivariance) $\Phi_B(\sigma \cdot \mathbf{s}) = \sigma \cdot \Phi_B(\mathbf{s})$.

Proof. (i) Superadditivity of floor: $\sum \lfloor s_i B \rfloor \leq \lfloor \sum s_i B \rfloor = \lfloor B \rfloor = B$. (ii) $B - \sum \lfloor s_i B \rfloor = \sum (s_i B - \lfloor s_i B \rfloor) = \sum \{s_i B\}$, a sum of $|A|$ fractional parts. (iii) $x \mapsto \lfloor xB \rfloor$ is nondecreasing. (iv) Coordinatewise application commutes with index permutation. $\square$

Remark 3.7 (Relation to classical apportionment). The Hamilton (largest-remainder) method redistributes the residual of (ii) to the largest fractional parts, achieving $\sum = B$ exactly; it pays with the Alabama paradox and population-monotonicity failures [3]. The deployed projection keeps the weaker guarantees (i)–(iv)—which are invariant under every monotone transformation of the inputs—and documents the residual rather than redistributing it. In the certified kernel the same discipline appears as *one-sided* truncation: saturating fixed-point division and multiplication lose value, never manufacture it, so flow conservation residuals are nonnegative by construction and were additionally *measured* across adversarial fixtures before being claimed.

3.4 Lanes, depth, and gauge structure

Priority lanes assign $\text{lane}(i) = \text{rank}(s_i) \bmod L$: a monotone map into the cyclic group $\mathbb{Z}/L\mathbb{Z}$ —an order-preserving projection of the Weyl chamber onto a cycle. Verification depth scales as $d_i = \min\left(D_{\max}, \max(1, \text{round}(D_{\max}(1 + s_i)/2))\right)$, an affine gauge of share mass. Neither carries new mathematics; both are recorded to fix notation for the conformance chapter.

Chapter 4

The Escort Cascade: Algebra and Multifractal Calculus

4.1 Masses and the partition function

Let each candidate carry masses $\mathbf{m}^{(k)} = (m_1^{(k)}, \dots, m_N^{(k)}) \in \mathbb{R}_{\geq 0}^N$ on $K = 4$ measure axes (option entropy, inverse cost, yield, salience (4.4) below). For a lens $q \in \mathbb{R}$ define the *partition function*

$$Z_k(q) = \sum_{i=1}^N \left(m_i^{(k)}\right)^q, \quad q \in \mathbb{R}, \quad (4.1)$$

finite for $q \geq 0$; for $q < 0$ it is defined iff all masses are positive (zero mass under a negative lens is a typed refusal, mirroring the kernel's ZeroMassUnderNegativeLens).

$$L_q^{(k)}(i) = \frac{\left(m_i^{(k)}\right)^q}{Z_k(q)}. \quad (4.2)$$

Proposition 4.1 (Limiting lenses). *L_0 is the uniform distribution (sibling-coverage convention: zero-mass siblings included); L_1 is proportional allocation; as $q \rightarrow +\infty$, $L_q \rightarrow e_{i^*}$ with $i^* = \arg \max m_i$ (exploitation); as $q \rightarrow -\infty$, $L_q \rightarrow e_{i_*}$ with $i_* = \arg \min_{m_i > 0} m_i$ (rarity). The compiled vocabulary $q \in \{2, 1, 0, -1\}$ samples all four regimes with one exponent each.*

4.2 Rényi dimensions and the lens calculus

The Rényi entropy of order q is $D_q(\mathbf{m}) = \frac{1}{1-q} \log Z(q)$ (with D_1 the Shannon entropy by limit), and the multifractal scaling exponent is $\tau(q) = (q-1)D_q = \log Z(q) - \log Z(1)$ when $\sum m_i = 1$.

Proposition 4.2 (Lens derivative identity). *For q in the interior of the convergence region and positive masses,*

$$\frac{d}{dq} \log Z(q) = \mathbb{E}_{L_q}[\log m], \quad \frac{d^2}{dq^2} \log Z(q) = \text{Var}_{L_q}(\log m) \geq 0. \quad (4.3)$$

Proof. Differentiate $\log Z = \log \sum m_i^q$: $\frac{d}{dq} \log Z = \frac{\sum m_i^q \log m_i}{Z} = \mathbb{E}_{L_q}[\log m]$. The second derivative is the variance of $\log m$ under L_q (standard exponential-family identity; $\log Z$ is convex in q by Hölder). $\square$

Corollary 4.3 (Monotone concentration). *Along the lens line, increasing q moves escort mass toward larger $\log m$ at a rate equal to its escort variance: the lens parameter is a concentration knob whose sensitivity is exactly the current dispersion of $\log m$. This is the quantitative form of the “exploitation” direction.*

4.3 The Gibbs identity for the deleted reference engine

The deleted softmax was $p_i(\tau) \propto e^{\tau S_i}$ with salience

$$S_i = \frac{\Omega_i Y_i}{\max(C_i, \varepsilon)}. \quad (4.4)$$

We record for the ledger’s historical chapter the identity that was once presented as a theorem:

Proposition 4.4 (Gibbs entropy identity). *For $p(\tau) \propto e^{\tau S}$ on a finite frontier,*

$$\frac{dH}{d\tau}(p(\tau)) = -\tau \operatorname{Var}_{p(\tau)}(S) \leq 0 \quad (\tau \geq 0), \quad (4.5)$$

with $H(p) = -\sum_i p_i \log p_i$.

Proof. $\frac{dp_i}{d\tau} = p_i(S_i - \mathbb{E}[S])$; hence $\frac{dH}{d\tau} = -\sum_i \frac{dp_i}{d\tau} (1 + \log p_i) = -\sum_i p_i (S_i - \mathbb{E}[S]) \log p_i$; substituting $\log p_i = \tau S_i - \log Z$ and expanding gives $-\tau \operatorname{Var}(S)$. $\square$

Remark 4.5 (Ledger entry). Proposition 4.4 holds for *every* Gibbs distribution over *any* score vector; it is an exponential-family fact, not a property of salience (4.4). It is printed here once, proved, and marked: the earlier presentation of such identities as characteristic theorems of the allocator was an audit finding (Ch. 11).

4.4 Legendre duality: the Hölder spectrum

Multifractal analysis pairs the lens line with the spectrum of local exponents. Define, for $\alpha \in \mathbb{R}$,

$$f(\alpha) = \inf_{q \in \mathbb{R}} \{ q\alpha - \tau(q) \}, \quad \alpha(q) = \frac{d\tau}{dq} = \mathbb{E}_{L_q}[\log m] + 1 \quad (4.6)$$

(the additive constant absorbs normalization). Then f is the Legendre–Fenchel transform of τ ; $f(\alpha)$ is the fractal dimension of the set of sites with local exponent α , and τ is recovered by the inverse transform. For the finite frontier, f is concave, supported on $[\log m_{\min}, \log m_{\max}]$.

Remark 4.6 (Why this matters for policy). A lens policy $(q_0, \dots, q_D)$ per cascade level is a *sampling of the τ -curve*. The Legendre picture says which local structures each lens sees: $q > 1$ sees the large-mass peaks (α small), $q < 1$ sees the rare support (α large), $q = 0$ sees everything uniformly. A compiled vocabulary $\{2, 1, 0, -1\}$ is a four-point quadrature of the entire multifractal spectrum—which is why Chapter 5 finds the blended policy flat *and* rarity-funding at once: the quadrature includes both endpoints.

4.5 Information geometry of the escort family

The map $q \mapsto L_q$ traces a curve in $\Delta_{>0}^{N-1}$ through the statistical manifold of the frontier. Under the Fisher metric $g_{ij} = \mathbb{E}_{L_q}[(\log m_i)(\log m_j)] - \mathbb{E}[\log m_i]\mathbb{E}[\log m_j]$ (the variance structure of Proposition 4.2), the escort curve is a geodesic up to reparameterization of Amari’s α -family with α affine in q [1, 23]: the lens line *is* the α -geometry of the frontier, traversed at unit speed proportional to escort variance. We do not fix the affine convention (authors differ); the structural statement is convention-free.

Remark 4.7. This is the precise sense in which the deleted softmax and the kept escort cascade are “the same mathematics”: both traverse the same geodesic, one with continuous parameter τ (requiring $e^{\tau S}$, a transcendental), one with integer parameter q (requiring only products m^q). Chapter 6 proves that this difference in arithmetic is the entire difference in governability.

Chapter 5

The Compiled Policy: Emergent Geometry of the Certified Blend

The deployed measure is not a single lens but the certified kernel’s blend over $K = 4$ measures and $Q = 4$ lenses with the compiled constants of Appendix B: lens registry $\mathbf{q} = (2, 1, 0, -1)$ and mixture matrix

$$\Lambda = \begin{pmatrix} \frac{2}{5} & \frac{3}{10} & \frac{1}{5} & \frac{1}{10} \\ \frac{1}{10} & \frac{1}{5} & \frac{3}{10} & \frac{2}{5} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{2} & 0 & \frac{1}{2} & 0 \end{pmatrix}, \quad \sum_q \Lambda_{kq} = 1 \quad \forall k, \quad \text{ETA} = \frac{1}{2}. \quad (5.1)$$

The kernel documents, and its correspondence tests pin, the blend identity

$$\text{allocate} = \sum_{k,q} \Lambda_{kq} \cdot \text{allocate_single_lens}(k, q), \quad (5.2)$$

where each `allocate_single_lens` is itself a full cascade under lens q : two-path flow routing with per-node split ρ_v , multiplicatively-weighted routing updates, and saturating Q16.16 arithmetic throughout. The interaction of these mechanisms makes the blend *emergent*: it is not a barycenter of textbook escorts, and this chapter is the mathematical record of what that means—including one conjecture that died for it.

5.1 The symmetric fixed point

Proposition 5.1 (Uniformity at symmetry). *When all N candidates present identical feature vectors, the kernel returns $s_i = 1/N$ exactly. Measured: for $N = 8$ and identical axes, each raw share is 0.125, exactly representable in Q16.16 (8192/65536), with raw total mass 1.0.*

Verification status. Symmetry of inputs forces every per-lens cascade to a permutation-invariant output; uniqueness of the fixed point follows from the deterministic integer arithmetic; the uniform vector is the only permutation-invariant distribution. Executed numerically to exactness; a formal proof against the Rust source is future work (§5.5). $\square$

5.2 A conjectured coverage floor, refuted

The natural first model of the blend is the *naive barycenter*: ignore routing and weights, treat each lens term as its pure escort π_{kq} , and read (5.2) as $s = \sum \Lambda_{kq} \pi_{kq}$. Since every $\pi_{k,0}$ (coverage) is uniform over N , this model predicts a floor:

Conjecture 5.2 (Naive coverage floor). $s_i \geq \frac{\Lambda_0}{N}$ for every candidate i , where $\Lambda_0 = \sum_k \Lambda_{k,0} = \frac{1}{5} + \frac{3}{10} + \frac{1}{4} + \frac{1}{2} = \frac{5}{4}$; for $N = 8$, $s_i \geq 5/32 = 0.15625$.

Theorem 5.3 (Refutation). *Conjecture 5.2 is false for the deployed kernel. Measured: with three real candidates (5.0, 0.1, 0.8, 0.4), (3.0, 0.1, 0.7, 0.21), (0.01, 0.01, 0.01, 10^{-6}) and five phantom pads, the dead-end raw share is $0.083679 < 5/32$, and the raw total mass is 0.4962, not $\sum_{k,q} \Lambda_{kq} = 4$: the phantom candidates absorb half the mass, and the blend is not the naive barycenter.*

Remark 5.4 (What the refutation teaches). The ρ -routing and multiplicative-weight machinery couple the lenses: the coverage component's uniform baseline is *routed* through the same flow splits as every other component and does not survive as an additive floor. The refutation is retained, per ledger discipline, as a first-class result: it delimits exactly which abstraction of the certified kernel is faithful (the blend identity (5.2) as an internal correspondence) and which is not (componentwise escort additivity).

Open Problem 5.5 (Sharp floor of the certified cascade). *Derive $\inf_{\mathbf{m}} s_i(\mathbf{m})$ over frontiers with $m_i \geq \varepsilon$, as a function of $(\Lambda, \mathbf{q}, \rho, \text{ETA})$, or prove no positive uniform floor exists once routing is coupled. A formalization of the kernel in a proof assistant, from which Proposition 5.1 and the floor both follow, is the natural instrument.*

5.3 Rarity funding: the measured anti-starvation mechanism

Model 5.6 (Rarity funding, measured). With $N = 8$ real candidates, seven at axes (1.0, 0.1, 0.5, 0.5) and one near-zero at (0.001, 0.001, 0.001, 10^{-9}): the *weak* candidate receives raw share 0.1721, exceeding the minimum *strong* share 0.1183. With one dominant candidate (10, 1, 2, 20) against seven weak ones, all weak candidates receive 0.0958 and the spike 0.3292: four orders of magnitude of mass ratio compress to a share ratio of 3.4.

The mechanism is the rare lens $q = -1$: under (4.2), m^{-1} amplifies small masses, so the blend's rare component concentrates on the weakest candidate—the compiled policy's built-in, quantitative anti-starvation. Together with extreme compression this gives the blend its measured character: *flat in the middle, protective at the bottom, compressed at the top*—a log-scale policy, matching the Legendre reading of §4.5 (the quadrature includes both endpoints of the Hölder spectrum).

5.4 A decision-theoretic reading (model)

Model 5.7 (Bayes hedging). Let the planner hold a Dirichlet posterior π over which branch is ultimately viable, with full support. Any policy assigning $s_i = 0$ forges the option value $\mathbb{E}_\pi[v_i \mathbf{1}\{i \text{ optimal}\}] > 0$. A policy with the measured floor behavior hedges at posterior-dependent cost; the flatness the performance lens reads as waste, the governance lens reads as the price of never being categorically wrong. This is a model of the *value* of the measured geometry, not a theorem about the kernel: the posterior, the value process, and the loss are assumptions.

Chapter 6

Arithmetic Determinism: Dyadic Rationals and the Transcendental Boundary

6.1 The dyadic grid

Q16.16 arithmetic computes over the nonnegative dyadic grid

$$\mathbb{D}_{16} = \left\{ k 2^{-16} : k \in \{0, \dots, 2^{32} - 1\} \right\} \subset \mathbb{Q}, \quad (6.1)$$

with saturating $+$, $-$, $\times$ and truncating $\div$ (floor toward zero). Every value is an exact rational; every operation is a total function on a finite totally ordered set; every result is bit-stable.

6.2 What IEEE 754 does and does not promise

IEEE 754 [9] specifies correctly-rounded $+$, $-$, $\times$, $\div$ and (since 2008) $\sqrt{}$; it does *not* specify transcendental functions, which may and do differ across libraries [7]. A softmax requires $\exp$; therefore a float softmax is not bit-portable in principle, only platform-coincident in practice. This is not a tuning problem; it is a classification problem: $\exp$ is transcendental, and no rounding rule makes its implementations agree.

6.3 The algebraization theorem

Theorem 6.1 (Algebraization of the exponential). *Let $m = e^S$ with S in the additive group generated by the feature logs. The escort family $\{L_q\}_{q \in \mathbb{R}}$ restricts on the integer sublattice $q \in \mathbb{Z}$ to maps computed from $\mathbf{m}$ by finite products and one exact division: $L_q(i) = m_i^q / \sum_j m_j^q$, m_i^q by $|q|$ saturating multiplications (and one reciprocal for $q < 0$). Consequently, on $\mathbb{D}_{16}$, every integer-lens escort is an algebraic function of the inputs—no transcendental is evaluated anywhere in the measure—whereas every non-integer lens (equivalently, every Gibbs softmax at $\tau \notin \{0, 1\}$ relative to the log-mass grid) requires one.*

Proof. Immediate from m^q for integer q being a finite product; division is specified exact-by-truncation; saturation is a monotone clamp into $\mathbb{D}_{16}$. The converse holds because $m^q = e^{qS}$ with qS leaving the log-grid almost surely forces evaluation of e^x at a non-grid point. $\square$

Theorem 6.2 (Bit-identity). *Any two faithful implementations of the same integer-lens cascade on the same dyadic grid produce identical output bits for identical inputs. Hence host- and language-independence of outputs is a structural property of the arithmetic, not a testing accident.*

Proof. Each operation is a specified total function on a finite set; composition of specified total functions is a specified total function; there is no latitude for an implementation to differ. $\square$

6.4 The executed proof: the cross-host golden fixture

Candidate	Axes $(\Omega, 1/C, Y, S)$	Share (exact)	Dyadic form
beta	$(2.0, 0.2, 0.8, 1.6)^\dagger$	0.2135009765625	13,983/65,536
alpha	$(4.0, 0.5, 0.5, 2.0)^\dagger$	0.153564453125	10,062/65,536
gamma	$(1.0, 0.1, 1.2, 0.4)^\dagger$	0.1290130615234375	8,452/65,536

Table 6.1: Golden fixture: one request through the identical artifact on native (CLI), CPython/`wasmtime`, and BEAM/`wasmex`; payloads byte-identical, values pinned in two languages. † Columns are $(\Omega, 1/C, Y, S)$ per the bridge’s axis order.

Theorem 6.2 predicts Table 6.1; the table executes the prediction. The failure mode of the old regime—the same request answered differently by different hosts—is not mitigated; it is unsatisfiable.

6.5 Determinism of the section

The canonical section σ_κ (Definition 3.1) supplies the second half of replay determinism: Proposition 3.2 (order-theoretic) plus Theorem 6.2 (arithmetic) give end-to-end bit-identical replay from unordered inputs to receipted plan.

Chapter 7

Conformance Geometry and Adversarial Separation

7.1 Event data as typed multigraphs

An OCEL 2.0 log is a typed multigraph $G = (E \sqcup O, \ell)$ with events E , objects O , and typed incidence $\ell \subseteq E \times \text{Qual} \times O$ [15]. Receipted allocations are events bound to plan/branch/-world objects; the log is the durable state of the governance process.

7.2 Alignments as geodesics

Let $\mathcal{M}$ be a process model (e.g. a discovered Petri net) and σ a trace. The synchronization product $\mathcal{S}(\sigma, \mathcal{M})$ is the weighted graph on pairs (log step, model step); the alignment cost

$$d(\sigma, \mathcal{M}) = \min_{\pi \in \Pi(\sigma, \mathcal{M})} \sum_{e \in \pi} w(e) \quad (7.1)$$

is the shortest-path (geodesic) distance in $\mathcal{S}$, computed by A^* search [17]. Conformance is the statement $d \leq \delta$; fitness the normalized ratio.

7.3 The counterfactual battery as adversarial neighborhood

Let $\mathcal{L} \subset \text{Traces}$ be the lawful language and $\mu : \mathcal{L} \rightarrow \mathcal{L}$ a mutator (premature actuation, unadmitted injection, skipped teardown, subtask inversion, crossed object links, allocation inversion). Governance holds when the detectors separate the mutant neighborhood from the lawful set:

$$\inf_{\sigma \in \mathcal{L}} d(\mu(\sigma), \mathcal{M}) \geq \delta_\mu > \sup_{\sigma \in \mathcal{L}} d(\sigma, \mathcal{M}). \quad (7.2)$$

Condition (7.2) is the metric form of fail-closed: a margin, not a verdict. The declared limitation stands: the mutator set is imagined by the same authors as the detectors

(Ch. 11), so (7.2) is verified on a designed neighborhood, and the margin against unimagined adversaries is unknown.

Chapter 8

Combinatorial Maximalism: The Sum–Product Law of Option Preservation

8.1 The sum–product law

Let a decision configuration be a graph whose nodes carry option sets $O_1, \dots, O_n$.

Principle 8.1 (Sum–product law). The configuration space grows with the *sum* of alternatives, $|G| = \Theta(\sum_i |O_i|)$, while the path space carries their *product*, $|\Pi| = \prod_i |O_i|$. A governed reasoner may hold all of G and must never enumerate Π : selection among preserved options proceeds by bounds and sections, with every truncation declared.

The Ehrhart count of Proposition 3.4 is the arithmetic instance: the budget polytope’s $\binom{B+N}{N}$ lattice points are bounded, never listed. The escort cascade is the dynamical instance: N candidates are all retained as support of L_q , and the lens line selects within them by a one-parameter family rather than by pruning to a singleton.

8.2 Option entropy as log-volume

Define the option state of a frontier as the pair $(\text{supp}(\mathbf{s}), H(\mathbf{s}))$. Since $H(\mathbf{s}) \leq \log |\text{supp}(\mathbf{s})|$ with equality iff uniform, entropy is the log-volume of the effective option set: the maximalist objective is precisely to keep log-volume of lawful alternatives bounded away from zero. Anti-starvation is the support invariant $|\text{supp}| \geq 1$ (projection law) and, in the compiled policy, the measured rarity-funding floor (§5): the two-level guarantee that the reasoner is never left with the empty product.

8.3 Pareto exposure, not argmax

An argmax returns one point of Δ^{N-1} : a decision with no residual structure. A governed allocation returns a point *with* its section (ordering), its lattice projection (Theorem 3.6),

its residual (declared), and its receipt—a full fiber of audit structure over the decision. Formally, the artifact exposes the map

$$\mathbf{s} \longmapsto \left(\sigma_{\kappa}(\mathbf{s}), \Phi_B(\mathbf{s}), B - \sum \Phi_B(\mathbf{s}), \text{SHA256}(\text{canon}(\mathbf{s})) \right), \quad (8.1)$$

and every consumer composes with this exposure rather than with $\mathbf{s}$ alone. This is the sum–product law at the interface level: the exposed structure grows additively with the laws enforced, while the space of compatible consumers grows multiplicatively.

8.4 Reversibility lattice

Options carry a partial order $a \sqsubseteq b$ (“ a is recoverable from b ’s consequences”). The design rule—COMPENSATABLE $\neq$ REVERSIBLE, and unknown reversibility is a fence—is the statement that the lattice’s order structure, not its labels, is the decision input: preference goes to the change preserving the most lawful upper bounds.

Chapter 9

The Phase Transition, Formally

9.1 Availability functionals

Model 9.1 (Integrity of multi-host agreement). Fix a per-host, per-change divergence probability p (platform arithmetic, version drift, silent fallback) and H hosts that must agree for an audit to pass. Per-implementation governance has integrity

$$I_{\text{multi}}(H) = (1 - p)^H \xrightarrow{H \rightarrow \infty} 0 \quad (\text{exponentially}), \quad (9.1)$$

while the single-kernel law replaces the H -fold conjunction with one fixture of constant cost: $I_{\text{kernel}}(H) = \mathbf{1}[\text{fixtures pass}]$, independent of H . This is the tube-reliability argument r^n of integrated electronics, transposed from components to hosts. It is a model: p is an assumed parameter, and independence across hosts is an assumption.

9.2 Failure-class removal as language emptiness

Let $\mathcal{B}(S)$ be the set of observable behaviors of system S . Old regime: there exist executions in Φ_{silent} (silent fallback), Φ_{diverge} (cross-host disagreement), $\Phi_{\text{unreceipted}}$ (actuation without content-bound trace), and these sets are nonempty and reached with positive probability. New regime: the executed language satisfies

$$\mathcal{B}(S) \cap (\Phi_{\text{silent}} \cup \Phi_{\text{diverge}} \cup \Phi_{\text{unreceipted}}) = \emptyset, \quad (9.2)$$

by construction: refusals are typed, transport equality is pinned by fixture, receipts are content-bound. Removal, not mitigation: the transition of Chapter 12 in set-theoretic form.

9.3 The order parameter

Define $\mu(t)$ = fraction of delivered product-surface lines that are hand-written rather than manufactured. The operating contract imposes the monotone descent law $\mu(t) \leq \mu(t - \Delta)$ per milestone; the session documented here moved μ discontinuously (the reference measure deleted outright), and the ledger (Ch. 11) is the trajectory of μ with receipts. In the

variational analogy, μ plays free energy: every audit step must lower it, and no step may trade a temporary increase for a promised future decrease without a paydown plan committed in the same change (Jaynes' maximum-entropy calculus applied to the codebase: the macrostate is the law surface; the microstates are its lawful renderings [10]).

Part III

Evidence and Argument

Chapter 10

The Governance Envelope, Summarized

The envelope formalized across Part II: *deterministic replay* (Props. 3.2, Thm. 6.2, Table 6.1); *conservation by construction* (Thm. 3.6 and the kernel’s measured one-sided residuals); *typed refusal* (limiting lenses, saturation semantics, transport refusals; metric form (7.2)); *receipt* (content-bound digests projected as OCEL 2.0 events). The mathematics is deliberately standard where it can be (escort families, Legendre duality, Ehrhart counting, apportionment) and novel exactly where the artifact is: the algebraization boundary (Thm. 6.1), the executed bit-identity, and the emergent blend geometry with its refuted conjecture and open floor.

Chapter 11

The Falsification Ledger

Deletion as a scholarly act: every claim adversarially audited; failed claims deleted with commit-anchored receipts; the hand-written surface monotone nonincreasing. The entries, in order of discovery:

1. **The circular benchmark:** comparator blind by construction to the evaluation metric, which was the allocator’s own objective; one comparator arm existed only in comments. Deleted as evidence; reborn as a characterization test with measured thresholds.
2. **The every-softmax theorems:** floor conservation, entropy monotonicity (Prop. 4.4), order preservation, permutation invarism—true, elementary, non-discriminating. Test file deleted (`c1d04ffc`).
3. **The notational equivalence:** $m = e^S$ turns $m^q / \sum m^q$ into a softmax; numerical agreement to 10^{-6} confirmed arithmetic, not a result. Deleted; replaced by the bit-identity fixture (real builds, real transports).
4. **The reference implementation:** the local float-softmax engine and its τ plumbing, deleted outright once the single-kernel law held; reintroduction pinned as a structural type error.
5. **The coverage-floor conjecture** (Conjecture 5.2): refuted by measurement during the writing of Chapter 5 (raw weak share $0.0837 < 5/32$), retained here as a result about the limits of barycentric abstraction for ρ -routed cascades.

The ledger converts the dissertation’s central risk—that a governance claim is itself unaudited—into structure: dishonesty would have required surviving an audit that deleted five of its own claims, including one mid-writing.

Chapter 12

The Phase-Change Analysis

12.1 A model of phase change

A technology undergoes a phase change when jointly: (1) a constraint *class* of the old regime is removed structurally (§9); (2) composition unlocks—assemblies of scale the old regime cannot reach at any price, because old failure rates grew with assembly size (Model 9.1); (3) a standard interface compounds improvement effort (steam: standardized power and threads; transistor: the planar process; here: the committed artifact with its provenance tripwires).

12.1.1 Steam

Early engines were thermally poor and economic only at the minehead; the phase change was removal of the biological/geographical power class, the factory system as composition unlock, and standardization as the compounding interface [11, 19].

12.1.2 The transistor

At demonstration, not faster than tubes; the initial win was removal of the filament failure class, making r^n assemblies arithmetically feasible, with integration as the compounding interface [4, 14].

12.2 The mapping

12.3 Disanalogies, stated

Scale: one laboratory, not an economy; mechanism, not history. **Binding constraint:** that auditability (not performance) binds consequential autonomy is a prediction, not a measurement. **Size of kernel:** $N \leq 8$ with a compiled lens vocabulary; scaling the pattern has named obstacles (lens-policy expressiveness, wasm artifact supply-chain governance, the known-mutation limitation of the battery).

	Old regime	This work
Constraint class removed	Divergent implementations; float non-portability; silent fallbacks; invisible decisions	One certified kernel; dyadic-grid replay (Thm. 6.2); typed refusal; receipted events
Composition unlock	Audit integrity $(1 - p)^H$ decays exponentially in hosts	Constant-cost fixture; any WASM host serves the identical law
Compounding interface	Tuning forks across variants	Artifact refresh with pin/fixture tripwires; ontology-declared policy
Legacy metric at transition	(Steam: efficiency; tubes: speed)	(Here: exploration sharpness—measured flatness and compression, Ch. 5)

Table 12.1: The structural mapping; the final row records the characteristic signature of transitions—worse on the old headline metric at the boundary.

12.4 Falsifiers

(i) per-implementation governance demonstrated to scale across heterogeneous hosts without a single-kernel discipline; (ii) surviving cross-host divergence of the certified kernel’s outputs (the fixtures fail and cannot be repaired by pin refresh); (iii) deployment evidence that operators refuse the guarantee at consequence scales; (iv) verification of arbitrary float policies made free, removing the constraint class by other means.

Chapter 13

Conclusion

The mathematics of this dissertation is the shadow of a deletion. When the local softmax died, what remained was structure: a simplex and its orbit space, a dyadic grid and its algebra, an escort line and its Legendre dual, a blend whose floor is open, a fixture whose equality is exact, and a ledger that records the conjectures killed in passing. The phase-change claim is now the precise statement that removing the implementation failure class converts host-count from an exponential liability into a constant, and the honest statement that everything else—adoption, scale, history—is future work with named falsifiers. Governance is not a property an artifact has; it is a property of the process allowed to modify the artifact. This dissertation has tried to be an instance of its own theory about it.

Appendix A

Artifact and Verification Map

Repositories: `seanchatmangpt/autofde-lab` (branch `feat/semantic-model-manufacturing`); sibling `ash_autofde` (local). Anchor commits: `39d244a9` vendoring; `465616a3` WASM transport; `3e2067f9` verification receipt; `c1d04ffc` reference deletion; `11e0559/1617520` BEAM host. Kernel vendored at `2ae60cf0`; artifact `wasm/artifacts/ bcinr_cmca_wasm.wasm` (421,988 bytes; ABI `cmca_alloc`, `cmca_free`, `cmca_rank_json`). Measurements of Ch. 5 were taken through `bcinr_wasm.call_rank` over the committed artifact; suites: 35 tests (`cmca`, `beam`), consumer suites green, `mix test` 10+1 green.

Appendix B

Compiled Constants

$N = 8$, $K = 4$, $Q = 4$; $\text{ETA} = 1/2$; lens registry $\mathbf{q} = (2, 1, 0, -1)$ (exploitation, proportional, coverage, rare); Λ as in (5.1) with exact entries $\{2/5, 3/10, 1/5, 1/10\}$, $\{1/10, 1/5, 3/10, 2/5\}$, $\{1/4\}^4$, $\{1/2, 0, 1/2, 0\}$; coverage column $\Lambda_0 = 5/4$. Values read from the vendored generated module at pin 2ae60cf0; the refuted floor used $\Lambda_0/N = 5/32$.

Appendix C

Measured Fixtures

Raw kernel outputs (before bridge renormalization), Q16.16-exact where marked: symmetric $N=8$: $0.125 = 8192/65536$ each, total 1.0. Three real (Ch. 5 refutation): (0.243393, 0.169144, 0.083679), total 0.4962. Seven strong + one weak: weak = $0.172088623046875 = 11280/65536$, strong min = $0.1182708740234375 = 7750/65536$. One spike + seven weak: weak 0.095825 each, spike 0.329163. Prime-budget conservation: $104,726 \leq 104,729$ over the full 8-branch frontier.

Appendix D

The Deleted Claims, For the Record

The headlines that did not survive audit: “mathematically rigorous and empirically sound” (one-line properties of any softmax); “proved and verified equivalence” (a change of notation); “empirical value analysis vs. alternatives” (a comparator in comments); “optimal collapse risk, maximal option preservation” (assertion); “every candidate retains mass $\geq \Lambda_0/N$ ” (refuted live, Ch. 5). Their deletion and refutation records are the citations. The reader is invited to distrust nothing less than everything else in this dissertation—and to check.

Bibliography

- [1] S.-i. Amari and H. Nagaoka, *Methods of Information Geometry*. AMS Translations of Mathematical Monographs 191, 2000.
- [2] P. Auer, N. Cesa-Bianchi, and P. Fischer, “Finite-time Analysis of the Multiarmed Bandit Problem,” *Machine Learning*, 47(2–4):235–256, 2002.
- [3] M. L. Balinski and H. P. Young, *Fair Representation: Meeting the Ideal of One Man, One Vote*. Yale University Press, 1982.
- [4] J. Bardeen and W. H. Brattain, “The Transistor, A Semi-Conductor Triode,” *Physical Review*, 74(2):230–231, 1948.
- [5] M. Beck and S. Robins, *Computing the Continuous Discretely: Integer-Point Enumeration in Polytopes*. Springer, 2007.
- [6] T. M. Cover and J. A. Thomas, *Elements of Information Theory*, 2nd edition. Wiley, 2006.
- [7] D. Goldberg, “What Every Computer Scientist Should Know About Floating-Point Arithmetic,” *ACM Computing Surveys*, 23(1):5–48, 1991.
- [8] T. C. Halsey, M. H. Jensen, L. P. Kadanoff, I. Procaccia, and B. I. Shraiman, “Fractal Measures and Their Singularities: The Characterization of Strange Sets,” *Physical Review A*, 33(2):1141–1151, 1986. †
- [9] IEEE, *IEEE Standard for Floating-Point Arithmetic*, IEEE 754-2008 (rev. 754-2019), 2008/2019.
- [10] E. T. Jaynes, “Information Theory and Statistical Mechanics,” *Physical Review*, 106(4):620–630, 1957.
- [11] D. S. Landes, *The Unbound Prometheus*. Cambridge University Press, 1969.
- [12] D. MacKenzie, *Mechanizing Proof: Computing, Risk, and Trust*. MIT Press, 2001.
- [13] J. Mokyr, *The Lever of Riches*. Oxford University Press, 1990.
- [14] G. E. Moore, “Cramming More Components onto Integrated Circuits,” *Electronics*, 38(8), 1965.
- [15] A. F. Ghahfarokhi, G. Park, A. Berti, and W. M. P. van der Aalst, “OCEL (Object-Centric Event Log) 2.0,” 2022. †

- [16] K. Küsters and W. M. P. van der Aalst, “Object-Centric Process Querying,” 2025. †
- [17] A. Berti, S. J. van Zelst, and W. M. P. van der Aalst, “Process Mining for Python (PM4Py),” 2019. †
- [18] A. Rényi, *Probability Theory*. North-Holland, 1970.
- [19] N. Rosenberg, *Perspectives on Technology*. Cambridge University Press, 1976.
- [20] C. Rosin, “Multi-Armed Bandits with Episode Context,” *Annals of Mathematics and Artificial Intelligence*, 61(3):203–230, 2011. †
- [21] C. E. Shannon, “A Mathematical Theory of Communication,” *Bell System Technical Journal*, 27:379–423, 623–656, 1948.
- [22] R. S. Sutton and A. G. Barto, *Reinforcement Learning: An Introduction*, 2nd edition. MIT Press, 2018.
- [23] C. Tsallis, “Possible Generalization of Boltzmann–Gibbs Statistics,” *Journal of Statistical Physics*, 52(1–2):479–487, 1988.

Bibliographic note. Entries marked † are provisional and must be verified against library records before deposit. Artifact citations (commits, constants, measurements) are exact and reproducible from the repository at the named commits.